INCH-POUND

MIL-DTL-46083D(MR)
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29 July 1988

DETAIL SPECIFICATION

ALUMINUM ALLOY ARMOR, EXTRUDED, WELDABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers extruded aluminum alloy armor shapes up to and including 2 inches in thickness (see 3.3 and 6.2).

1.2 Classification. Material furnished under this specification should be one of the following classes (see 6.2):

Class I - 5083 and 5456
Class II - 7039 and 2219

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, ATTN: AMSRL-WM-M, Aberdeen Proving Ground, MD 21005-5069 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 – V_{50} Ballistic Test for Armor

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service (DAPS), Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI H35.2 Dimensional Tolerances for Aluminum Mill Products (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B557 Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products (DoD adopted)

ASTM E34 Chemical Analysis of Aluminum and Aluminum-Base Alloys (DoD adopted)

ASTM E227 Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique (DoD adopted)

ASTM E607 Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique, Nitrogen Atmosphere (DoD adopted)

ASTM E716 Sampling Aluminum and Aluminum Alloys for Spectrochemical Analysis

ASTM E1251 Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Argon Atmosphere, Point-to-Plane, Unipolar Self-Initiating Capacitor Discharge (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2) and before production has commenced, samples of the specified item shall be made available to the contracting officer or his authorized representative for approval

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in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be manufactured by the process proposed for use on production.

3.1.1 Change in mill processing. After an armor material has successfully complied with the requirements of this specification, any deliberate change in the mill processing by the manufacturer shall be drawn to the attention of the procuring activity and the contractor. If the material processing is changed, the first article tests of this specification may be again called upon to assure compliance with the requirements of this specification.

3.2 Chemical composition. The chemical composition limits for class I and class II materials shall be within the limits shown in table I (see 6.2).

TABLE I. Chemical composition limits, (weight percent).

Class	Alloy	Silicon max.	Iron max.	Copper	Manganese	Magnesium	Chromium	Zinc	Titanium	Others, max. Each	Others, max. Total	Aluminum
I	5083	0.40	0.40	0.10a/	0.40-1.0	4.0-4.9	0.05-0.25	0.25a/	0.15a/	0.05	0.15	Rem.
	5456	0.25	0.40	0.10a/	0.50-1.0	4.7-5.5	0.05-0.20	0.25a/	0.20a/	0.05	0.15	Rem.
II	7039	0.30	0.40	0.10a/	0.10-0.40	2.3-3.3	0.15-0.25	3.5-4.5	0.10a/	0.05	0.15	Rem.
	2219	0.20	0.30	5.8-6.8	0.20-0.40	0.02a/	--	0.10a/	0.02-0.10	b/	0.15	Rem.

a/Maximum
b/Zirconium 0.10-0.25
Vanadium 0.05-0.15
Others each 0.05 max.

3.3 Mechanical properties. Unless otherwise specified (see 6.2), the mechanical properties of class I and class II extrusions shall meet the requirements of table II, when tested in accordance with 4.6.2.

TABLE II. Mechanical properties.a/

Class	Direction of specimen	Thickness, inches	Tensile strength min. psi	Yield strength 0.2% offset min. psi	Elongation min. percent
I	Longitudinal	Up to 2.000, incl	45,000	35,000	9
II	Longitudinal	Up to 2.000, incl	57,000	48,000	8

a/ See 6.5 and 6.6

3.4 Ballistic requirements. Each ballistic test extrusion shall be proof fired for resistance to penetration by determining a protection ballistic limit, BL(P), at 0° obliquity. The minimum required ballistic limit for class I materials shall be determined from the values shown in tables A-I, A-II and A-III. The minimum required ballistic limit for class II materials shall be determined from the values shown in tables A-IV, A-V, A-VI and A-VII.

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3.5 Thermal processing (class 1 only). After acceptance testing, any thermal processing exceeding 250°F (121°C) performed on the extrusion excluding fabrication by welding, will necessitate revivification for conformance to 3.3 and 3.4. Deviation from this requirement shall be subject to review and approval by the procuring activity.

3.6 Dimensions. Dimensions shall be as shown in the drawing specified in the invitation for bids, contract or order (see 6.2).

3.6.1 Tolerances. Unless otherwise specified in the contract or order (see 6.2), the extrusions submitted for acceptance shall not vary from the specified dimensions shown in ANSI H35.2.

3.7 Marking for identification. Each extrusion shall be marked in at least one location in characters not less than 3/8 inch high with the following information. Impression stamping shall not be used, except in areas designated by the drawing (see 6.2).

a. Manufacturer's name, trademark or both.
b. Number of this specification.
c. Class and alloy designation.
d. Part or drawing number.
e. Lot number.

3.7.1 Ballistic test extrusions. Ballistic test extrusions for first article and acceptance plates shall be marked with the information in 3.7. In addition, first article extrusions shall be marked "PRE" and acceptance test extrusions marked "ACC". This marking shall be impression stamped in at least two opposing locations in letters not less than 1/2-inch high.

3.8 Ballistic test extrusion information. A properly executed armor data report shall be submitted containing the information as specified in MIL-STD-662.

3.9 Workmanship. Extruded shapes shall be clean, smooth, uniform in quality as well as free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect their use in fabrication within the limits of good commercial practices.

3.9.1 Surface and edge condition. Surface cracks, edge cracks or edge laminations shall be cause for rejection.

4. VERIFICATION

4.1 Clarification of inspections. The inspection requirements specified herein are classified as conformance inspection (see 4.2)

4.2 Conformance inspection. Test for acceptance of individual lots consist of testing all requirements specified in section 3. Verification conformance inspection is in accordance with the provisions herein. Ballistic tests shall be performed at a government test activity selected by the procuring activity (see 6.2).

4.3 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3.1).
b. Conformance (acceptance) inspection (see 4.3.2).

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4.3.1 First article inspection. First article inspection shall consist of the following:

a. Chemical analysis (see 3.2 and 4.7.1).
b. Mechanical properties (see 3.3 and 4.7.2).
c. Ballistic tests (see 3.4 and 4.7.3).
d. Dimensions (see 3.6 and 4.6.2).
e. Identification marking (see 3.7, 3.7.1 and 4.6.1).
f. Workmanship (see 3.9 and 4.6.1).

4.3.2 Conformance (acceptance) inspection. Lot acceptance inspection shall consist of the following:

a. Chemical analysis (see 3.2 and 4.7.1).
b. Mechanical properties (see 3.3 and 4.7.2).
c. Ballistic tests (see 3.4 and 4.7.3).
d. Dimensions (see 3.6 and 4.6.2).
e. Identification marking (see 3.7, 3.7.1 and 4.6.1)
f. Workmanship (see 3.9 and 4.6.1).
g. Preparation for shipment (see section 5).

4.4 Lot. A lot shall consist of finished extrusions, of the same class, shape, dimensions, alloy and temper, which are submitted for inspection as a unit, at essentially the same time and have a total weight of 25,000 pounds or less.

4.5 Sampling.

4.5.1 First article testing.

a. Chemical composition. Samples for chemical analyses shall be removed from each extrusion selected for ballistic test. A certification of the chemical composition of the alloy shall be furnished with the test extrusions.

b. Mechanical properties. One tension test specimen for first article shall be removed from each extrusion that has been selected for ballistic test. Location and type of specimen shall be as specified in 4.7.2.

c. Ballistic tests. At least two extruded samples having sufficient area for performance of the ballistic test shall be submitted for each designated thickness and extruded shape shown on the drawing and which the manufacturer has under contract. Because of the variety of possible shapes and sizes, the manufacturer must contact the ballistic test agency to determine the minimum number needed to satisfactorily conduct the ballistic tests.

4.5.2 Acceptance lots.

a. Chemical composition, ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured from the same source of molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

b. Chemical composition, finished product analysis. When sampling has not been made in accordance with 4.5.2 a., one sample shall be randomly taken for each 4,000 pounds (or less) of every lot in accordance with ASTM E716 or ASTM E34.

c. Mechanical properties. Unless otherwise specified (see 6.2) from material having a nominal weight of less than one pound per lineal foot, one tension test sample shall be selected from each inspection lot weighing l,000 pounds or less; from inspection lots weighing more than 1,000 pounds, one

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additional sample shall be taken for each 1,000 pounds or fraction thereof in excess of the first 1,000 pounds. From material having a nominal weight of one pound or more per lineal foot, one tension-test sample shall be taken from each inspection lot consisting of 1,000 feet, or less; from inspection lots consisting of more that 1,000 feet, one additional sample shall be taken from each 1,000 feet or fraction thereof in excess of the first 1,000 feet. Only one tension-test specimen shall be taken from any one piece when more that one piece is available.

d. Ballistic tests. Because of the variety of possible shapes and sizes the manufacturer must contact the ballistic test agency to determine the minimum number of extrusions per lot which are needed to conduct the ballistic test(s).

4.6 Examination.

4.6.1 Visual. Each extrusion shall be examined to determine compliance with the identification marking (see 3.7) and workmanship (see 3.9) requirements.

4.6.2 Dimensional. One extrusion from each lot (see 4.4) shall be measured to determine compliance with the requirements for dimensions and tolerances (see 3.6 and 3.6.1).

4.6.3 Preparation for shipment. Prior to shipment, examination shall be made to determine compliance with section 5.

4.7 Tests.

4.7.1 Chemical composition. Chemical analysis shall be conducted in accordance with ASTM methods E34, E227, E607, or E1251.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. The specimens shall be taken from the extruded shape in the longitudinal direction. From shapes of nominal thickness of 0.75 to 1.50 inches, inclusive, tension test specimens shall be taken with the axis midway between the extruded surfaces; from shapes of nominal thickness greater than 1.50 inches, the axis of the test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Ballistic testing. Ballistic testing shall be in accordance with appendix A. Test extrusions shall be tested at locations specified on the drawing or in the contract (see 6.2). Extrusions shall be tested in accordance with the thickness ranges shown in table III to determine the projectile as well as the required V_{50} protection ballistic limit (see A.5.1.2). Extruded shape thickness as measured by the ballistic test agency shall be used to determine the required ballistic limit. Individual thickness measurements are read to the nearest 0.001 inch and the average of these readings reported to the nearest 0.005 inch. Thickness measurements of the extruded shape shall be made of the areas to be ballistically tested.

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TABLE III. Acceptance ballistic test extrusions.

Class	Nominal design thickness specified on the drawing (inches)	Projectile a/
I	0.750 - 1.000	Caliber .50 FSP
	1.001 - 2.000	Caliber .30 AP, M2
	1.001 - 1.700	20 mm FSP
II	0.750 - 0.950	Caliber .50 FSP
	0.951 - 1.500	Caliber .30 AP, M2
	0.951 - 1.500	20 mm FSP
	1.501 - 2.000	Caliber .50 AP, M2

a/ FSP denotes fragment simulating projectile and AP M2 denotes armor piercing model 2 projectile.

4.8 Test specimens.

4.8.1 Rejection. Unless otherwise specified (see 6.2) where one or more test specimens fail to meet the requirements of the specification the lot represented by the specimen or specimens shall be subject to rejection.

4.8.2 Retest. When no sampling plan is provided or approved by the procuring agency (see 6.2) and where there is evidence that indicates that the specimen was not representative of the lot of material, and when the detail specification does not otherwise specify, at least two specimens shall be selected to replace each test specimen which failed. All specimens so selected for retest shall meet the requirements of the specification or the lot shall be subject to rejection.

4.8.3 Ballistic. Rejection and retest of ballistic test extrusions shall be in accordance with A.5.2.

4.9 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced, and providing the manufacturer agrees to maintain the same manufacturing procedures.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The aluminum alloy armor covered by this specification is military unique. The armor will be used in military vehicles to protect the operator and passengers against small arms fire and shrapnel.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Nominal thickness of aluminum armor plate (see 1.1).
c. Type, class, and alloy (see 1.2).
d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.3 and A.2.1.1).
e. When first article is required (see 3.1).
f. Chemical composition and mechanical properties of aluminum armor (see 3.2 and 3.3).
g. Drawings, dimensions, and tolerances (see 3.6 and 3.6.1).
h. Marking for identification (see 3.7).
i. Destination of tension test samples (see 4.2).
j. Locations of tension test specimens (see 4.5.2 c).
k. Locations for determination of ballistic limits (see 4.7.3).
l. Rejection requirement, if other than in 4.8.1.
m. Retest requirement, if other than in 4.8.2.
n. Packaging requirements (see 5.1).

6.2.1 Armor material test reports. When this specification or specification requirements are used in an acquisition and data are required to be delivered, the data requirements identified in 3.8 should be developed and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. The data specified should be delivered by the Contractor in accordance with the contract or purchase order requirements.

6.3 Ownership of ballistic test extrusions.

6.3.1 First article and acceptance test extrusions. First article and acceptance test extrusions that comply with the requirements of this specification are considered as part of the lot of extrusions that they represent, and ownership of them passes to the Government with acceptance of that lot. First article and acceptance test extrusions that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the manufacturer as does the rejectable lot which they represent.

6.3.1.1 Disposal instructions for tested extrusions. At the time of shipment of the first article and acceptance test extrusions the proving ground should be informed by the manufacturer, in writing, as to the disposition of first article and acceptance test which fail the test. If instructions for disposal of the tested extrusions are not received after 15 days, the proving ground will exercise the prerogative of disposing of the material.

6.4 Stress corrosion. To prevent failure in service due to stress corrosion, care should be exercised in design and fabrication to avoid stressing this material in the short transverse direction. No alkaline etching should be permitted.

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6.5 Mechanical to ballistic requirements. The minimum mechanical properties specified (see 3.3) may not assure aluminum armor extrusions meeting the specified ballistic requirements (see 3.4).

6.6 Metric units. When required the units for inch, foot, pound, pounds per square inch, and feet per second may be converted to the metric equivalents by multiplying them by the following conversion factors:

English unit	Multiplied by	Equals	Metric equivalent
inch	0.025400	=	meter (m)
foot	0.304800	=	meter (m)
pound (lb_m)	0.453592	=	kilogram (kg)
pound per square inch (psi)	0.006895	=	megapascal (MPa)
feet per second (ft/s)	0.304800	=	meter per second, (m/s)

6.7 Definitions.

6.7.1 Contracting officer. The term "contracting officer" means the person executing a contract on behalf of the Government, and any other officer or civilian employee who is properly designated contracting officer; and the term includes, except as otherwise provided, the authorized representative of a contracting officer acting within the limits of his authority.

6.7.2 Contractor. The term “contractor” is defined as the organization having a direct contract with the procuring activity.

6.7.3 Manufacturer. The term “manufacturer” is defined as the organization actually performing the operations covered by this specification.

6.7.4 Procuring activity. The term "procuring activity" is that activity of the Government which actually initiates the request for procurement and maintains the records of the procurement.

6.8 Subject term (key word) listing.

Ballistic requirements
Obliquity
Projectile
Thermal processing

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

BALLISTIC TESTING OF ALUMINUM ALLOY ARMOR, EXTRUDED, WELDABLE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor, extruded, weldable, when tested in accordance with the provisions of this specification. When there is a mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 – V_{50} Ballistic Test for Armor

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service (DAPS), Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact should be considered fair when an unyawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any impact which is not a complete penetration should be considered a partial penetration.

A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

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APPENDIX A

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required ballistic limit shall be in accordance with the values shown in tables A-I through A-VII. In those cases where the BL(P) is within ± 10 fps of the minimum required value for the measured average thickness (to the nearest 0.005 inch), an interpolation of the appropriate table will be performed. The average plate thickness, computed to the nearest 0.001 inch, will be used to determine the minimum required ballistic limit.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V_{50} Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature conditioning. Prior to the test, the test item(s) will be temperature conditioned for at least eight hours. Thermostatic control will be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) will be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data will be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit will be computed on the basis of 4 or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing will continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits will be reported as such on the armor data report in accordance with MIL-STD-662.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, will be fired to reduce the gap to 25 fps or less. The ballistic limit will then be recomputed using the above criteria. The recomputed BL(P) will be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete penetration velocity should result in a better evaluation of the BL(P).

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APPENDIX A

A.5.2 Rejection and retest of ballistic extrusions.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or order, failure of any of the first article test extrusions to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest extrusions shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest extrusions shall be submitted for first article testing, and both tests must pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or order, failure of a test extrusion to meet the ballistic requirements indicates failure of the lot; however, the final decision will depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test extrusion representing a lot fails to meet the ballistic requirement, immediately upon notification of the failure of the test extrusion, the manufacturer may, at his own expense, submit two additional extrusions for ballistic test. If either one or both of these extrusions fail the test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test extrusions. If either or both of the extrusions from a retreated lot fail, the lot shall be permanently rejected. Ballistic retest extrusions shall be marked "R1" and "R2" respectively in addition to the requirements of 3.7.

A.5.3 Disposition of ballistic test extrusions.

A.5.3.1 First article test extrusions. Upon request of the applicant within 15 days after ballistic testing, first article extrusions will be returned "as is" to the applicant, at his expense, unless the extrusions were destroyed in testing.

A.5.3.2 Acceptance test extrusions. Acceptance test extrusions that comply with the requirements of this specification are considered as part of the lot they represent, and ownership of the test extrusions passes to the Government with the acceptance of that lot. Acceptance test extrusions that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as the rejectable lot does. The failed extrusions will be returned, upon request, as in A.5.3.1.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits for class I armor extrusions (protection at 0° obliquity) - caliber .30 AP M2 projectile.

Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps
0.970	1749	1.185	1991	1.400	2208
0.975	1755	1.190	1997	1.405	2213
0.980	1761	1.195	2002	1.410	2218
0.985	1767	1.200	2007	1.415	2223
0.990	1773	1.205	2012	1.420	2228
0.995	1779	1.210	2018	1.425	2232
1/ 1.000	1784	1.215	2023	1.430	2237
1.005	1790	1.220	2028	1.435	2242
1.010	1796	1.225	2033	1.440	2247
1.015	1802	1.230	2039	1.445	2251
1.020	1808	1.235	2044	1.450	2256
1.025	1814	1.240	2049	1.455	2261
1.030	1819	1.245	2054	1.460	2265
1.035	1825	1.250	2059	1.465	2270
1.040	1831	1.255	2064	1.470	2275
1.045	1837	1.260	2069	1.475	2279
1.050	1843	1.265	2075	1.480	2284
1.055	1848	1.270	2080	1.485	2289
1.060	1854	1.275	2085	1.490	2293
1.065	1860	1.280	2090	1.495	2298
1.070	1865	1.285	2095	1.500	2303
1.075	1871	1.290	2100	1.505	2307
1.080	1877	1.295	2105	1.510	2312
1.085	1882	1.300	2110	1.515	2317
1.090	1888	1.305	2115	1.520	2321
1.095	1893	1.310	2120	1.525	2326
1.100	1899	1.315	2125	1.530	2330
1.105	1904	1.320	2130	1.535	2335
1.110	1910	1.325	2135	1.540	2339
1.115	1916	1.330	2140	1.545	2344
1.120	1921	1.335	2145	1.550	2348
1.125	1927	1.340	2150	1.555	2353
1.130	1932	1.345	2155	1.560	2358
1.135	1937	1.350	2160	1.565	2362
1.140	1943	1.355	2165	1.570	2367
1.145	1948	1.360	2170	1.575	2371
1.150	1954	1.365	2174	1.580	2376
1.155	1959	1.370	2179	1.585	2380
1.160	1965	1.375	2184	1.590	2384
1.165	1970	1.380	2189	1.595	2389
1.170	1975	1.385	2194	1.600	2393
1.175	1981	1.390	2199	1.605	2398
1.180	1986	1.395	2204	1.610	2402

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APPENDIX A

TABLE A-I. Minimum required ballistic limits for class I armor extrusions (protection at 0° obliquity) - caliber .30 AP M2 projectile - Continued.

Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps
1.615	2407	1.755	2528	1.895	2644
1.620	2411	1.760	2532	1.900	2648
1.625	2416	1.765	2536	1.905	2652
1.630	2420	1.770	2540	1.910	2656
1.635	2424	1.775	2545	1.915	2660
1.640	2429	1.780	2549	1.920	2664
1.645	2433	1.785	2553	1.925	2668
1.650	2438	1.790	2557	1.930	2672
1.655	2442	1.795	2561	1.935	2676
1.660	2446	1.800	2566	1.940	2680
1.665	2451	1.805	2570	1.945	2684
1.670	2455	1.810	2574	1.950	2688
1.675	2459	1.815	2578	1.955	2692
1.680	2464	1.820	2582	1.960	2696
1.685	2468	1.825	2586	1.965	2700
1.690	2472	1.830	2590	1.970	2704
1.695	2477	1.835	2595	1.975	2708
1.700	2481	1.840	2599	1.980	2712
1.705	2485	1.845	2603	1.985	2715
1.710	2490	1.850	2607	1.990	2719
1.715	2494	1.855	2611	1.995	2723
1.720	2498	1.860	2615	2/ 2.000	2727
1.725	2502	1.865	2619	2.005	2731
1.730	2507	1.870	2623	2.010	2735
1.735	2511	1.875	2627	2.015	2739
1.740	2515	1.880	2631	2.020	2743
1.745	2519	1.885	2635	2.025	2747
1.750	2524	1.890	2640	2.030	2751

1/ Specification requirements begin with 1.001 inch ordered thickness. Required BL(P) = 1785 fps.

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits for class I armor extrusions (protection at 0° obliquity) - 20 mm FSP projectile.

Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps
0.970	1310	1.185	1663	1.400	2101
0.975	1317	1.190	1672	1.405	2112
0.980	1325	1.195	1681	1.410	2124
0.985	1332	1.200	1690	1.415	2135
0.990	1340	1.205	1700	1.420	2147
0.995	1347	1.210	1709	1.425	2158
1/ 1.000	1355	1.215	1718	1.430	2170
1.005	1362	1.220	1728	1.435	2182
1.010	1370	1.225	1737	1.440	2193
1.015	1378	1.230	1747	1.445	2205
1.020	1385	1.235	1756	1.450	2217
1.025	1393	1.240	1766	1.455	2229
1.030	1401	1.245	1776	1.460	2241
1.035	1409	1.250	1785	1.465	2253
1.040	1417	1.255	1795	1.470	2265
1.045	1424	1.260	1805	1.475	2277
1.050	1432	1.265	1815	1.480	2290
1.055	1440	1.270	1825	1.485	2302
1.060	1448	1.275	1835	1.490	2314
1.065	1456	1.280	1845	1.495	2327
1.070	1465	1.285	1855	1.500	2339
1.075	1473	1.290	1865	1.505	2352
1.080	1481	1.295	1875	1.510	2364
1.085	1489	1.300	1885	1.515	2377
1.090	1497	1.305	1895	1.520	2390
1.095	1506	1.310	1906	1.525	2403
1.100	1514	1.315	1916	1.530	2416
1.105	1522	1.320	1927	1.535	2429
1.110	1531	1.325	1937	1.540	2442
1.115	1539	1.330	1948	1.545	2455
1.120	1548	1.335	1958	1.550	2468
1.125	1556	1.340	1969	1.555	2481
1.130	1565	1.345	1979	1.560	2494
1.135	1574	1.350	1990	1.565	2508
1.140	1582	1.355	2001	1.570	2521
1.145	1591	1.360	2012	1.575	2535
1.150	1600	1.365	2023	1.580	2548
1.155	1609	1.370	2034	1.585	2562
1.160	1618	1.375	2045	1.590	2575
1.165	1627	1.380	2056	1.595	2589
1.170	1636	1.385	2067	1.600	2603
1.175	1645	1.390	2078	1.605	2617
1.180	1654	1.395	2089	1.610	2631

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APPENDIX A

TABLE A-II. Minimum required ballistic limits for class I armor extrusions (protection at 0° obliquity) - 20 mm FSP projectile - Continued.

Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps
1.615	2645	1.655	2760	1.695	2880
1.620	2659	1.660	2775	2/ 1.700	2895
1.625	2673	1.665	2789	1.705	2910
1.630	2688	1.670	2804	1.710	2926
1.635	2702	1.675	2819	1.715	2941
1.640	2716	1.680	2834	1.720	2957
1.645	2731	1.685	2849	1.725	2973
1.650	2745	1.690	2864	1.730	2988

1/ Specification requirements begin with 1.001 inch ordered thickness. Required BL(P) = 1356 fps.

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits for class I armor extrusions (protection at 0° obliquity) - caliber .50 FSP projectile.

Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps
0.700	1578	0.870	2140
0.705	1592	0.875	2160
0.710	1607	0.880	2179
0.715	1621	0.885	2198
0.720	1636	0.890	2218
0.725	1651	0.895	2238
0.730	1666	0.900	2258
0.735	1681	0.905	2278
0.740	1696	0.910	2298
0.745	1711	0.915	2318
1/ 0.750	1727	0.920	2339
0.755	1742	0.925	2360
0.760	1758	0.930	2381
0.765	1774	0.935	2402
0.770	1790	0.940	2423
0.775	1806	0.945	2445
0.780	1823	0.950	2466
0.785	1839	0.955	2488
0.790	1856	0.960	2510
0.795	1872	0.965	2532
0.800	1889	0.970	2555
0.805	1906	0.975	2577
0.810	1923	0.980	2600
0.815	1941	0.985	2623
0.820	1958	0.990	2646
0.825	1975	0.995	2670
0.830	1993	2/ 1.000	2693
0.835	2011	1.005	2717
0.840	2029	1.010	2741
0.845	2047	1.015	2765
0.850	2066	1.020	2790
0.855	2084	1.025	2814
0.860	2103	1.030	2839
0.865	2121		

1/ Specification requirements begin with this ordered thickness.

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits for class II armor extrusions (protection at 0° obliquity) - caliber .50 FSP projectile.

Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps
0.720	1700	0.855	2267
0.725	1718	0.860	2291
0.730	1737	0.865	2315
0.735	1756	0.870	2340
0.740	1774	0.875	2365
0.745	1794	0.880	2390
1/ 0.750	1813	0.885	2415
0.755	1832	0.890	2441
0.760	1852	0.895	2467
0.765	1872	0.900	2493
0.770	1892	0.905	2519
0.775	1912	0.910	2546
0.780	1933	0.915	2573
0.785	1954	0.920	2600
0.790	1975	0.925	2628
0.795	1996	0.930	2656
0.800	2017	0.935	2684
0.805	2039	0.940	2712
0.810	2060	0.945	2741
0.815	2082	2/ 0.950	2770
0.820	2105	0.955	2799
0.825	2127	0.960	2829
0.830	2150	0.965	2858
0.835	2173	0.970	2889
0.840	2196	0.975	2919
0.845	2219	0.980	2950
0.850	2243		

1/ Specification requirements begin with this ordered thickness.

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits for class II armor extrusions (protection at 0° obliquity) - 20 mm FSP projectile.

Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps
0.920	1137	1.125	1525	1.330	2028
0.925	1146	1.130	1535	1.335	2042
0.930	1154	1.135	1546	1.340	2056
0.935	1162	1.140	1557	1.345	2070
0.940	1171	1.145	1568	1.350	2084
0.945	1179	1.150	1579	1.355	2099
1/ 0.950	1188	1.155	1590	1.360	2113
0.955	1197	1.160	1602	1.365	2128
0.960	1205	1.165	1613	1.370	2142
0.965	1214	1.170	1624	1.375	2157
0.970	1223	1.175	1636	1.380	2172
0.975	1232	1.180	1647	1.385	2187
0.980	1240	1.185	1659	1.390	2202
0.985	1249	1.190	1670	1.395	2217
0.990	1258	1.195	1682	1.400	2232
0.995	1267	1.200	1694	1.405	2247
1.000	1277	1.205	1706	1.410	2263
1.005	1286	1.210	1717	1.415	2278
1.010	1295	1.215	1729	1.420	2294
1.015	1304	1.220	1742	1.425	2310
1.020	1314	1.225	1754	1.430	2325
1.025	1323	1.230	1766	1.435	2341
1.030	1333	1.235	1778	1.440	2357
1.035	1342	1.240	1791	1.445	2373
1.040	1352	1.245	1803	1.450	2390
1.045	1361	1.250	1816	1.455	2406
1.050	1371	1.255	1828	1.460	2422
1.055	1381	1.260	1841	1.465	2439
1.060	1391	1.265	1854	1.470	2455
1.065	1401	1.270	1867	1.475	2472
1.070	1411	1.275	1880	1.480	2489
1.075	1421	1.280	1893	1.485	2506
1.080	1431	1.285	1906	1.490	2523
1.085	1441	1.290	1919	1.495	2540
1.090	1451	1.295	1932	2/ 1.500	2557
1.095	1461	1.300	1946	1.505	2575
1.100	1472	1.305	1959	1.510	2592
1.105	1482	1.310	1973	1.515	2610
1.110	1493	1.315	1986	1.520	2628
1.115	1503	1.320	2000	1.525	2645
1.120	1514	1.325	2014	1.530	2663

1/ Specification requirements begin with 0.951 inch ordered thickness. Required BL(P) = 1190 fps.

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits for class II armor extrusions (protection at 0° obliquity) - caliber .30 AP M2 projectile.

Thickness Inches	Required BL (P),fps	Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL (P), fps
0.920	1906	1.125	2153	1.330	2376
0.925	1912	1.130	2159	1.335	2382
0.930	1919	1.135	2165	1.340	2387
0.935	1925	1.140	2170	1.345	2392
0.940	1931	1.145	2176	1.350	2397
0.945	1938	1.150	2182	1.355	2402
1/ 0.950	1944	1.155	2187	1.360	2407
0.955	1950	1.160	2193	1.365	2412
0.960	1957	1.165	2199	1.370	2418
0.965	1963	1.170	2204	1.375	2423
0.970	1969	1.175	2210	1.380	2428
0.975	1975	1.180	2215	1.385	2433
0.980	1981	1.185	2221	1.390	2438
0.985	1988	1.190	2226	1.395	2443
0.990	1994	1.195	2232	1.400	2448
0.995	2000	1.200	2237	1.405	2453
1.000	2006	1.205	2243	1.410	2458
1.005	2012	1.210	2248	1.415	2463
1.010	2018	1.215	2254	1.420	2468
1.015	2024	1.220	2259	1.425	2473
1.020	2030	1.225	2265	1.430	2478
1.025	2036	1.230	2270	1.435	2483
1.030	2042	1.235	2276	1.440	2488
1.035	2048	1.240	2281	1.445	2493
1.040	2054	1.245	2286	1.450	2498
1.045	2060	1.250	2292	1.455	2503
1.050	2066	1.255	2297	1.460	2508
1.055	2072	1.260	2303	1.465	2513
1.060	2078	1.265	2308	1.470	2518
1.065	2084	1.270	2313	1.475	2523
1.070	2090	1.275	2319	1.480	2528
1.075	2096	1.280	2324	1.485	2533
1.080	2101	1.285	2329	1.490	2537
1.085	2107	1.290	2334	1.495	2542
1.090	2113	1.295	2340	2/ 1.500	2547
1.095	2119	1.300	2345	1.505	2552
1.100	2125	1.305	2350	1.510	2557
1.105	2130	1.310	2355	1.515	2562
1.110	2136	1.315	2361	1.520	2567
1.115	2142	1.320	2366	1.525	2570
1.120	2148	1.325	2371	1.530	2576

1/ Specification requirements begin with 0.951 inch ordered thickness. Required BL(P) = 1190 fps.

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VII. Minimum required ballistic limits for class II armor extrusions (protection at 0° obliquity) - caliber .50 AP M2 projectile.

Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps	Thickness Inches	Required BL(P), fps
1.470	1865	1.660	2008	1.850	2142
1.475	1869	1.665	2012	1.855	2146
1.480	1872	1.670	2015	1.860	2149
1.485	1876	1.675	2019	1.865	2153
1.490	1880	1.680	2023	1.870	2156
1.495	1884	1.685	2026	1.875	2159
1/ 1.500	1888	1.690	2030	1.880	2163
1.505	1892	1.695	2033	1.885	2166
1.510	1896	1.700	2037	1.890	2170
1.515	1900	1.705	2041	1.895	2173
1.520	1903	1.710	2044	1.900	2176
1.525	1907	1.715	2048	1.905	2180
1.530	1911	1.720	2051	1.910	2183
1.535	1915	1.725	2055	1.915	2186
1.540	1919	1.730	2059	1.920	2190
1.545	1922	1.735	2062	1.925	2193
1.550	1926	1.740	2066	1.930	2197
1.555	1930	1.745	2069	1.935	2200
1.560	1934	1.750	2073	1.940	2203
1.565	1938	1.755	2076	1.945	2207
1.570	1941	1.760	2080	1.950	2210
1.575	1945	1.765	2083	1.955	2213
1.580	1949	1.770	2087	1.960	2217
1.585	1953	1.775	2090	1.965	2220
1.590	1956	1.780	2094	1.970	2223
1.595	1960	1.785	2097	1.975	2226
1.600	1964	1.790	2101	1.980	2230
1.605	1968	1.795	2104	1.985	2233
1.610	1971	1.800	2108	1.990	2236
1.615	1975	1.805	2111	1.995	2240
1.620	1979	1.810	2115	2/ 2.000	2243
1.625	1982	1.815	2118	2.005	2246
1.630	1986	1.820	2122	2.010	2249
1.635	1990	1.825	2125	2.015	2253
1.640	1993	1.830	2129	2.020	2256
1.645	1997	1.835	2132	2.025	2259
1.650	2001	1.840	2135	2.030	2263
1.655	2004	1.845	2139		

1/ Specification requirements begin with 1.501 inch ordered thickness. Minimum required BL(P) = 1889 fps.

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

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4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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8. PREPARING ACTIVITY

a. NAME US Army Research Laboratory Weapons and Material Research Dirctorate	b. TELEPHONE *Include Area Code)* (1) Commercial (410)306-0725 (2) AUTOVON 458-0725
c. ADDRESS *(Include Zip Code)Commander* ARL/WMRD ATTN; AMSRL/WM-M Aberdeen Proving Ground, MD 21005-5069	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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